

Computer Networks Notes

1. Introduction to Computer Networks

A computer network is a group of interconnected devices that communicate and share resources.

2. Types of Networks

- LAN (Local Area Network)
- MAN (Metropolitan Area Network)
- WAN (Wide Area Network)

3. Network Devices

- Router: Connects different networks.
- Switch: Connects devices in a LAN.
- Hub: Broadcasts data to all devices.
- Repeater: Regenerates signals.

4. OSI Model

- Physical Layer
- Data Link Layer
- Network Layer
- Transport Layer
- Session Layer
- Presentation Layer
- Application Layer

5. TCP vs UDP

TCP is connection-oriented and reliable.

UDP is faster but connectionless.

6. IP Address

An IP address uniquely identifies a device on a network.

7. Network Topologies

- Star
- Bus
- Ring
- Mesh

8. Protocols

Protocols define communication rules between devices.

Examples:

- HTTP
- HTTPS
- FTP
- SMTP